

Class **Contact:**

Contains attributes:

String name

String phone number

String email address

String address

String birthday

String notes

Contains methods:

Constructers.

Setters/getters

Class **LinkedList:**

Contains attributes:

Node<Type> head

Node<Type> current

Contains methods:

Method findFirst ()

input: none

results: first element set as the current element.

output: none

Method findNext ()

input: none

results: element following the current element is made the current element.

output: none.

Method retrieve (Type e)

input: none

results: current element is copied into e.

output: element e.

Method update (Type e)

input: e

results: the element e is copied into the current node.

output: none.

Method insert (Type e)

input: e.

results: a new node containing element e is created and inserted after the current element in the list. The new element e is made the current element.

If the list is empty e is also made the head element.

output: none.

Method remove ()

input: none.

results: the current element is removed from the list.

If the resulting list is empty current is set to NULL.

If successor of the deleted element exists it is made the new current element otherwise first element is made the new current element.

Method full (boolean flag)

input: none.

results: if the number of elements in L has reached the maximum number allowed then flag is set to true otherwise false.

output: flag.

Method empty (boolean flag)

input: none.

results: if the number of elements in L is zero, then flag is set to true otherwise false.

Output: flag.

Method last (boolean flag)

input: none.

Results: if the last element is the current element then flag is set to true
otherwise false.

Output: flag.

Class Phonebook:

Contains attributes:

LinkedList<Contact> contactList

LinkedList<Event> eventList

Contains methods:

Method addContact (String name, String phone number, String email address, String address, String birthday, String notes)

Input: String name, String phone number, String email address, String address,

String birthday, String notes.

Result: checks if a contact already exists with the same name or phone if not it should create a contact with said information and adds it via insert method in the linked list while being sorted via name. if contact is added then return true otherwise return false.

Output: flag.

Method searchContact(String identifier)

Input: String identifier (name or phone number)

Result: searches the contact list for a contact with a name or

A phone number that matches the identifier and then makes it the current

Returns true if its found otherwise returns false

Output: flag.

Method printContacts(String identifier)

Input: String identifier (email or address or birthday)

Result: searches the list for all contact with an email address or

An address or a birthday that matches the identifier and then prints them all it

Output: none.

Method removeContact(String identifier)

Input: String identifier (name or phone number)

Result: searches the list via searchContact with the mentioned identifier

And then removes it via remove method in linked list and returns true otherwise return false.

After that the method should check event list for an event with the same identifier via searchEvent method and removes it as well.

Output : flag.

Method printContactByFirstName(String name)

Input: String name.

Result: prints all contact's details with the same input name.

Output: none.

Method addEvent (String title, String date, String time, String location, String contactName)

Input: String title, String date, String time, String location, String contactName

Result: checks if contact exist then checks if no conflict in time then creates a new Event with said info and then inserts it while being sorted alphabetically.

Should return true if add otherwise return false.

Output: flag.

Method searchEvent(String name)

Input: String name

Result: searches the event list for an event with a contact with the matching name or and then makes it the current Returns true if its found otherwise returns false

Output: flag.

Method printAllEvents()

Input: none.

Result: prints all events with their details. Output: none.

Class **Event**:

Contains attributes:

String title

String date

String time

String location

String contactName

Contains methods:

Constructers.

Setters/getters